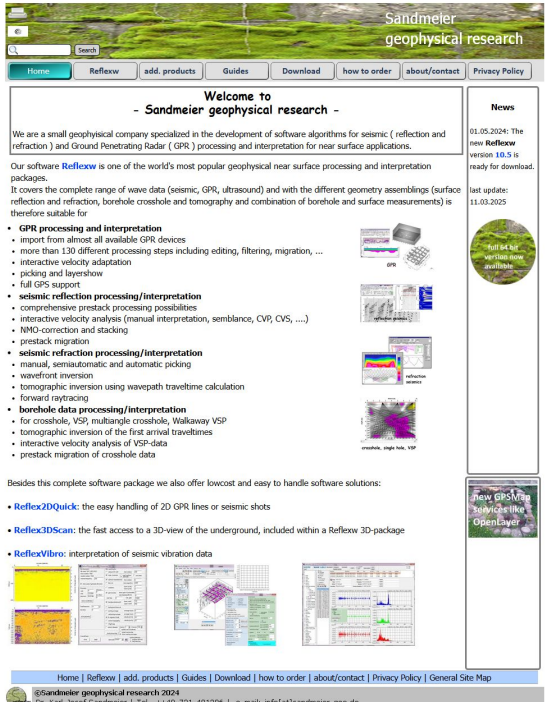
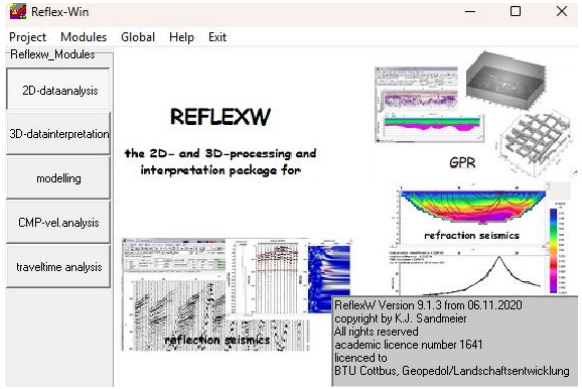
Reflexw



Manual and Videos →
https://www.sandmeier-geo.de/index.html



Our guide →
https://docs.google.com/document/d/1-pYSXHyyq7VpLRZ5x-Jyaaif0vGlBCSb9d42LRsN3Jw/edit?usp=sharing

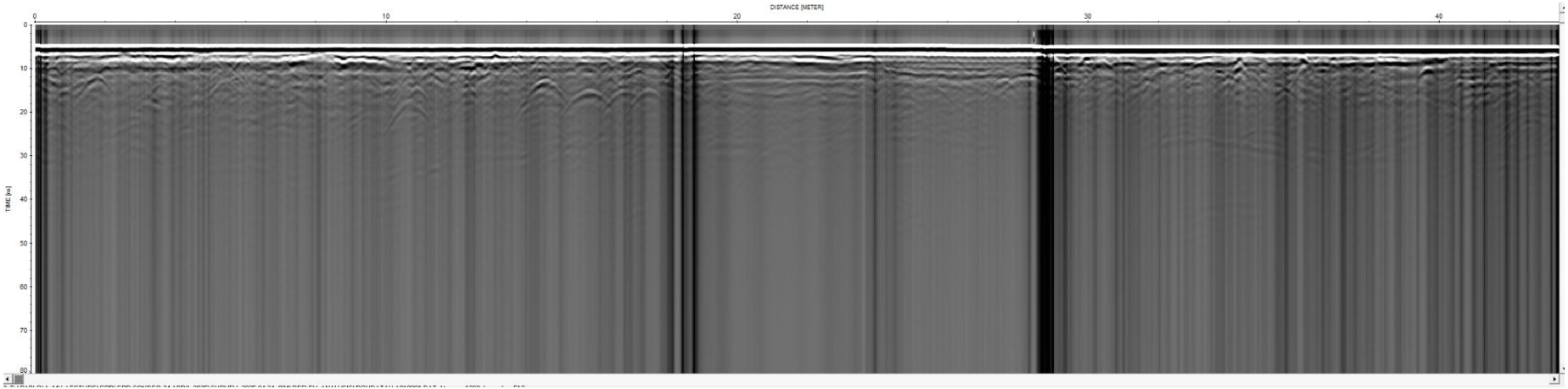
Data processing

1) **Get familiar with ReflexW**: basic steps include running HD Converter, creating project folders, loading data, importing profiles. Then explore the basics of *plot options*.

2) **Apply filters**: progressively apply processing filters to remove noise and enhance reflections.

3) **Initial visual analysis**: look at the profile globally, identify major features, distinguish between the two frequencies, and compare with the map generated in the field.

What we see from the start…



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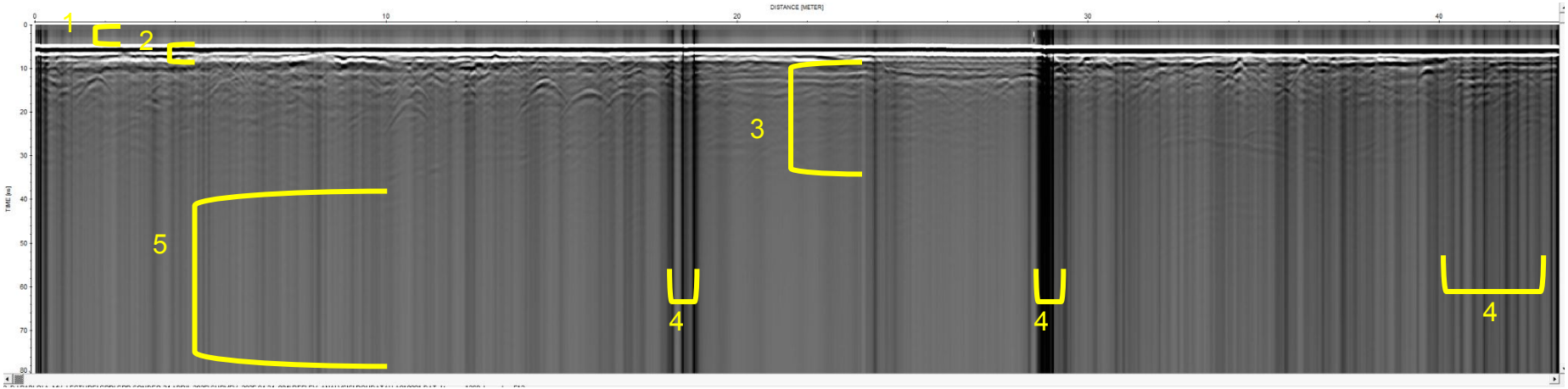

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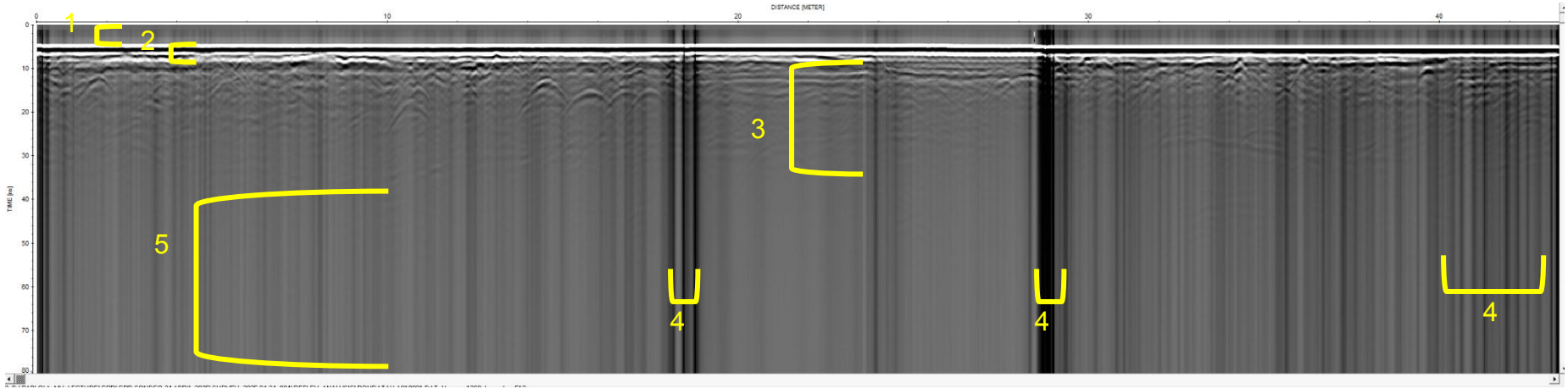
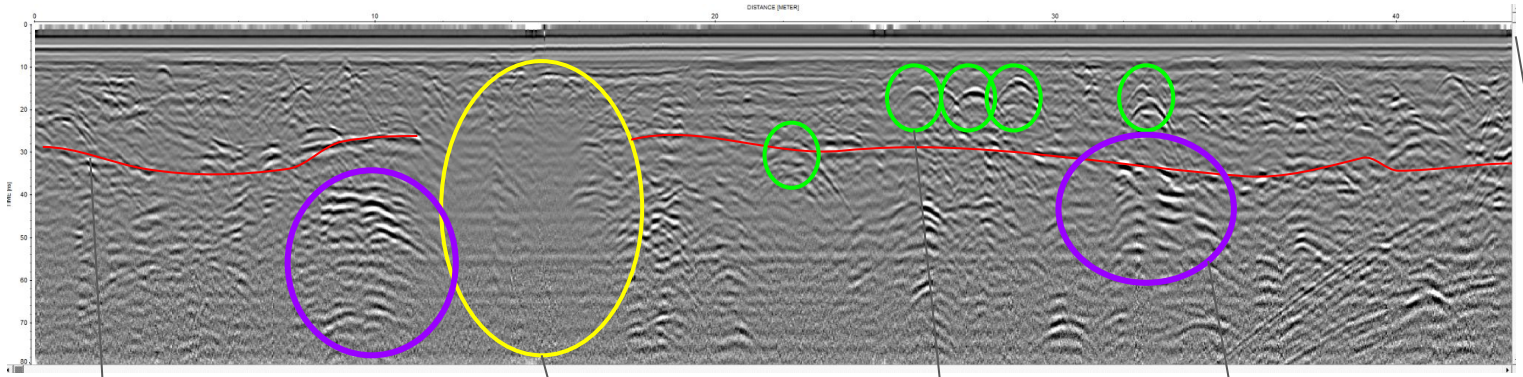


1) **Pre direct wave zone** → air gap or time before any signal arrives
2) **DIrect wave** → Signal travelling directly from the transmitter to the receiver → remove with background removal or muting
3) **Subsurface reflections** → interfaces layers, buried objects (pipes rocks etc)
4) **Vertical noise** → this is cahused by adquisition issues and or hardware interference. These are not real subsurface objects → spike removal
5) **Deep noise** → difffuse and irregular reflections; caused by signal attenuation

And after the processing…



interface:
fill to natural soil?

Homogeneous or low constrast material:
Maybe compacted fill without objects?
or uniform dry sand?

Point objects, buried structures:
pipes, rocks, metallic elements

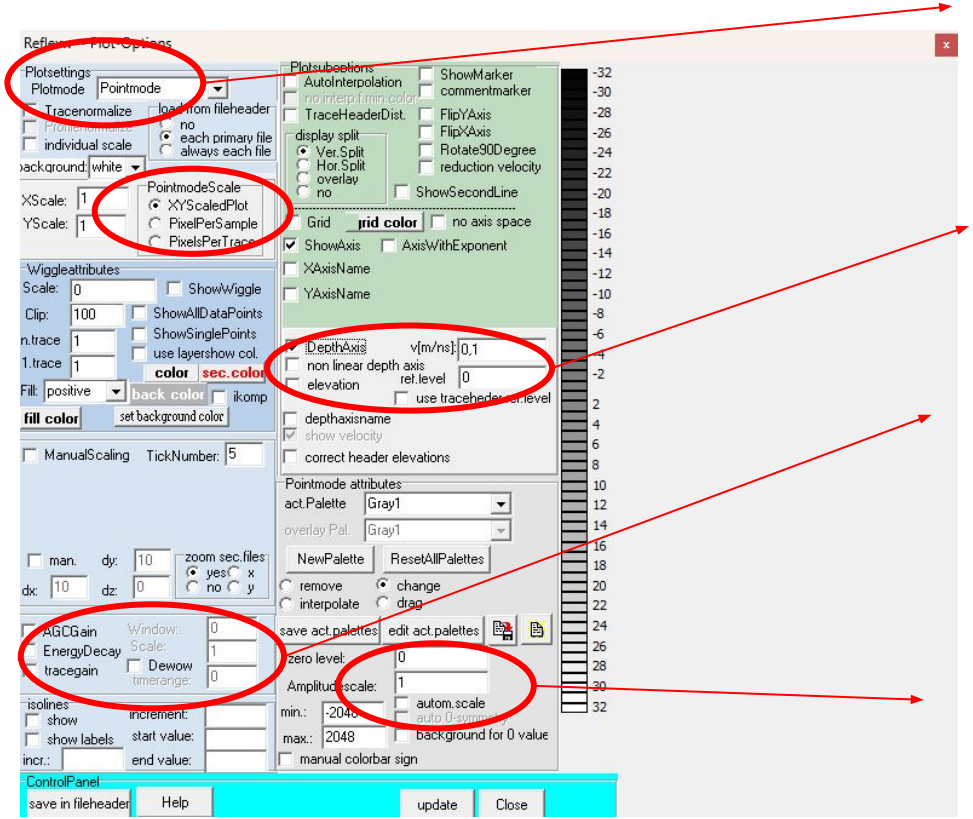
construction remains?
metallic elements?
compacted material?

Direct signal and contact with pavement

Start working with the data.

(You can follow the guide)

Getting familiar with ReflexW:



Plotsettings:

- Choose between Wiggle mode (traces as wiggles) and Radargram mode (amplitude as grayscale).

Depth Axis:

- Set the propagation velocity (v) to convert time (ns) to depth (m). Adjusting the velocity according to the subsurface material.

Basic Filters:

- Energy Decay: compensation for the loss of signal energy with depth. We use it for visualizing deeper reflections
- Dewow: Removes low-frequency noise. Essential for getting clearer reflections.

Amplitude Scale:

- Adjusts the visual intensity of the traces : enhance weak reflections or avoid saturation

Depth Axis

- The velocity value we are going to enter in ReflexW for DepthAxis is used just to convert the vertical time axis to depth.
- If there are multiple materials, it is common to use an average value representative of the dominant material.

$$propagation\ speed = \frac{c}{\sqrt{\varepsilon_r}}$$

$$depth = \frac{velocity * time}{} = \frac{propagation\ speed * range}{2} = \frac{c}{\sqrt{\varepsilon_r}} * \frac{range}{2}$$

$$c = 30\ cm/ns$$

- Air $\epsilon_r = 1\ v \simeq 30\ cm/ns$
- Dry Sand $\epsilon_r = 3\ to\ 6\ \ v \simeq 12.2\ cm/ns\ to\ 17.3\ cm/ns$
- Wet Sand $\epsilon_r = 20\ to\ 30\ \ v \simeq 5.5\ cm/ns\ to\ 6.7\ cm/ns$
- Dry Soil $\epsilon_r = 4\ to\ 6\ \ v \simeq 12.2\ cm/ns\ to\ 15\ cm/ns$
- Wet Soil $\epsilon_r = 10\ to\ 15\ \ v \simeq 7.7\ cm/ns\ to\ 9.5\ cm/ns$
- Fresh Water $\epsilon_r = 80\ \ v \simeq 3.4\ cm/ns\ to$
- Concrete $\epsilon_r = 6\ to\ 12\ \ v \simeq 8.7\ cm/ns\ to\ 12.2\ cm/ns$
- Asphalt $\epsilon_r = 4\ to\ 10\ \ v \simeq 9.5\ cm/ns\ to\ 15\ cm/ns$

Setting an incorrect speed results in depth errors:

- Too high speed: Depth is overestimated
- Too low speed: Depth is underestimated

Dewow Filter: Removing Low-Frequency Noise

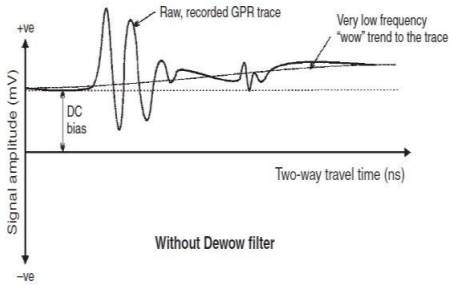
The filter is used to eliminate very low-frequency components that are not part of the actual reflections.

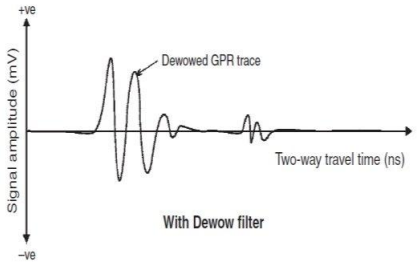
Why do we need to filter

1. **Noise from the equipment (DC noise):**
   - Electronic systems generate slow baseline drift and low-frequency noise.
   - These components appear as long, smooth oscillations or a curved baseline in the radargram.

2. **Low-frequency components within the pulse spectrum:**
   - A 200 MHz antenna does not emit a pure 200 MHz tone but a broadband pulse.
   - The emitted signal contains spectral components both below and above 200 MHz.
   - Dewow may remove very low-frequency parts of the pulse spectrum, and it wont affect useful signals if the window size is appropriate.

$$f_c = \frac{1}{2T}$$

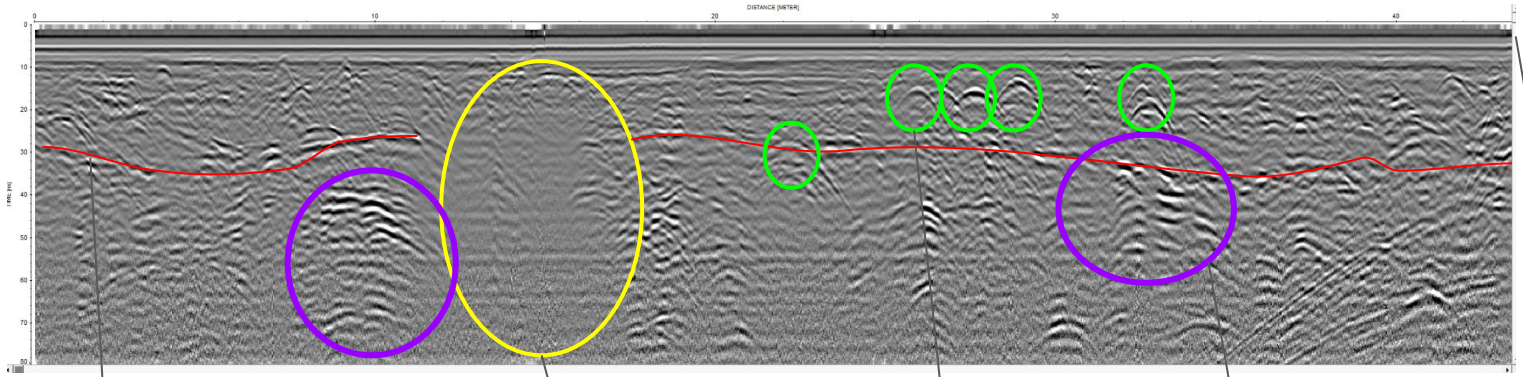




Antenna freq	Dewow time window (aprox)	Cut off freq (aprox)
200 MHz	10-20 ns	50-25 MHz
600 MHz	2-5 ns	250-100 MHz

Interpretation

600 Mhz antenna - just after modifying dewow, gain, and plot scale



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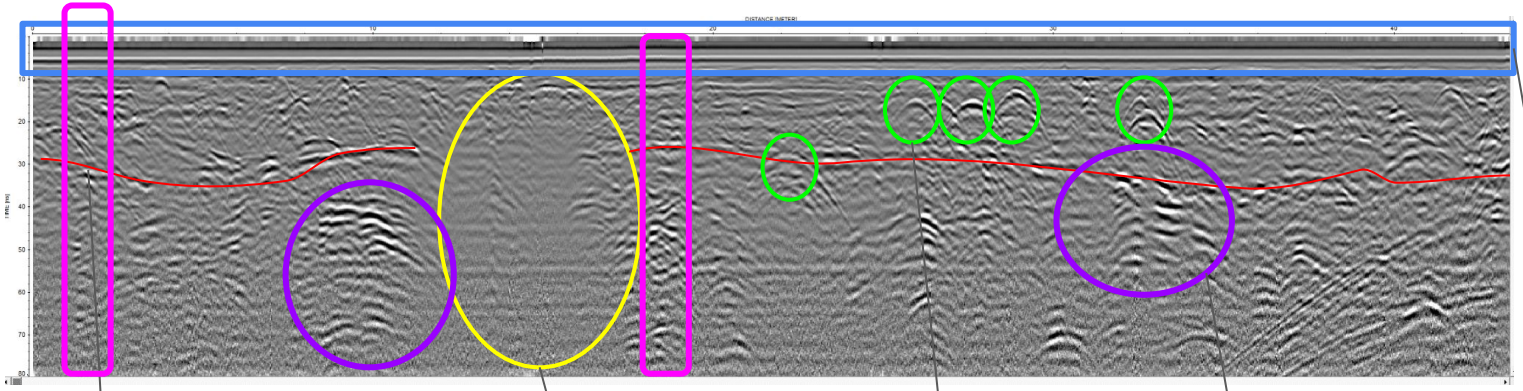

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Data processing

Data import and initial setup

- Load raw data and se correct parameters (units, time window)

Basic processing

- Flip (in case is necessary)
- Static correction: Align the time zero to account fo signal delay

Amplitude and gain adjustments

- Amplitude scale. You can adjust it for optimal visual contrast
- Energy decay. This compensates for amplitude loss

Noise reduction and filtering

- Background removal. This subtracts average trace to eliminate systematic noise
- Band pass filter. Keep only the frequencies you want within a useful range.

Advanced processing

- Migration. this corrects hyperbolic reflection to their true position
- Deconvolution. Increase resolution by compressing the wavelet.

DO NOT overprocess the data!